Navigating Austrian Citizenship Independently

A Data Analysis of Time, Cost, and Bureaucratic Complexity

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Introduction

This case study presents a data-driven analysis of the Austrian citizenship application process based on a real applicant experience. The purpose of the project is to identify key barriers related to time, cost, and procedural complexity and to translate those findings into practical recommendations for improving the process for independent applicants.

The structure of this case study follows the same six-phase framework used in the Bellabeat case study: **Ask, Prepare, Process, Analyze, Share, and Act**. That framework requires a clear business task, a description of the data source, documentation of data cleaning and manipulation, a summary of analysis, supporting visualizations and key findings, and high-level recommendations.

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1. Ask

Business task

The business task of this case study is to analyze the Austrian citizenship application process in order to identify the main barriers faced by a fully independent applicant and determine how the process could be made more transparent, efficient, and accessible.

Problem statement

The Austrian citizenship process is often perceived as difficult, highly bureaucratic, and time-intensive. For an applicant completing the process independently, the challenge is not

only meeting the formal legal requirements, but also navigating unclear documentation requirements, cross-border dependencies, waiting periods and financial obligations.

Guiding questions

This case study addresses the following questions:

1. Which stages of the process take the most time?
2. Which requirements generate the highest costs?
3. Which steps are perceived as the most difficult?
4. What relationship exists between time, cost and complexity?
5. What improvements could make the process more accessible for future applicants?

Stakeholders

The key stakeholders in this analysis are:

- independent citizenship applicants
- Austrian administrative authorities
- policy makers
- institutions involved in documentation, legal verification and citizenship processing

Why this matters

The insights from this case study can help reveal where the process creates unnecessary burden and where targeted reforms could improve fairness, usability, and efficiency.

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2. Prepare

Data source

The primary dataset used in this project was **self-collected** and documents the full Austrian citizenship process from pre-application preparation to final approval. The dataset contains **17 structured steps** and includes the following variables:

- Step ID
- Phase
- Category
- Step Description
- Time (days)
- Cost (€)
- Difficulty (1–5)
- Notes

In addition to the self-collected dataset, the project was informed by the official Austrian citizenship document checklist, which outlines the types of records, financial documentation, translations, language proof, and other official requirements applicants may need to provide.

Data organization

The dataset was structured as a single table in Excel and later used in Python and SQLite. Each row represents one application step, while each column represents one variable. This structure made it possible to conduct descriptive analysis, group data by category and phase, and also compare time, cost and difficulty across the process.

Data credibility and limitations

The dataset is original and experience-based, which makes it highly relevant to the case study question. At the same time, it has clear limitations:

- it reflects one applicant's experience rather than a large population
- some values, especially difficulty scores, are interpretive rather than externally verified
- it captures a real process in depth, but not multiple applicant profiles

Even with these limitations, the dataset is useful for identifying process bottlenecks, structural inefficiencies and applicant pain points.

Privacy and accessibility

The dataset was kept at the level of process steps and did not include sensitive personal identifiers. This allowed the process to be analyzed as a system without exposing private information.

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3. Process

Tools used

This project used three main tools:

- **Excel** for initial data entry, cleaning, and chart building
- **Python (pandas)** for data exploration, grouped summaries, and exported analysis files
- **DB Browser for SQLite** for SQL-based querying, summary statistics, and grouped analysis

This combination reflects the workflow encouraged in the Bellabeat roadmap, which recommends using spreadsheets, SQL, or Python to import, explore, clean, transform, and summarize data.

Data cleaning and preparation

The dataset was cleaned manually in spreadsheet software before being analyzed further. The following cleaning steps were performed:

- standardized all time values into days
- standardized all cost values into euros
- reviewed step names for consistency
- assigned a difficulty score on a scale from 1 to 5
- reviewed the table for missing or inconsistent values
- structured the data into a single worksheet for easier analysis

Python workflow

Python and pandas were used to inspect and summarize the dataset. The main functions used were:

- `pd.read_excel()` to import the spreadsheet
- `head()` to preview the first rows
- `info()` to inspect column structure and data types
- `describe()` to generate summary statistics
- `isnull().sum()` to identify missing values
- `pd.to_numeric()` to standardize numeric columns
- `groupby()` to aggregate time, cost, and difficulty by category and phase
- `sort_values()` to identify the longest, most expensive, and most difficult steps
- `to_csv()` to export summary tables for further analysis

A simplified version of the Python workflow is shown below:

```
import pandas as pd

df = pd.read_excel("citizenship.xlsx")

print(df.head())
print(df.info())
print(df.describe())
print(df.isnull().sum())

df.columns = [
    "step_id",
    "phase",
    "category",
    "step_description",
    "time_days",
    "cost_eur",
    "difficulty",
    "notes"
```

```
]

df["time_days"] = pd.to_numeric(df["time_days"], errors="coerce")
df["cost_eur"] = pd.to_numeric(df["cost_eur"], errors="coerce")
df["difficulty"] = pd.to_numeric(df["difficulty"], errors="coerce")

time_by_category = df.groupby("category")["time_days"].sum().reset_index()
cost_by_category = df.groupby("category")["cost_eur"].sum().reset_index()
difficulty_by_category = df.groupby("category")["difficulty"].mean().reset_index()

print(time_by_category)
print(cost_by_category)
print(difficulty_by_category)
```

SQL workflow

SQL was used in DB Browser for SQLite to explore the same dataset using queries. After importing the CSV file into SQLite, a cleaned table was created and used to generate summary statistics and grouped views. The main SQL tasks followed the Bellabeat-style approach of importing data, exploring rows and distinct values, calculating summary statistics, investigating trends, and saving results into summary tables.

```
SELECT COUNT(*) AS total_rows

FROM citizenship_clean;


SELECT DISTINCT phase

FROM citizenship_clean;


SELECT DISTINCT category

FROM citizenship_clean;


SELECT

    SUM(time_days) AS total_time_days,

    SUM(cost_eur) AS total_cost_eur,

    AVG(difficulty) AS avg_difficulty
```

```
FROM citizenship_clean;


SELECT
    category,
    SUM(time_days) AS total_time_days
FROM citizenship_clean
GROUP BY category
ORDER BY total_time_days DESC;


SELECT
    category,
    SUM(cost_eur) AS total_cost_eur
FROM citizenship_clean
GROUP BY category
ORDER BY total_cost_eur DESC;


SELECT
    phase,
    SUM(time_days) AS total_time_days
FROM citizenship_clean
GROUP BY phase
ORDER BY total_time_days DESC;
```

Verification

The dataset was checked in all three environments:

- visually in Excel
- through summary statistics in Python

- through row counts and grouped queries in SQL

This cross-checking helped ensure consistency before interpretation.

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4. Analyze

Overview

After preparing and cleaning the dataset, a descriptive analysis was conducted to identify patterns related to time, cost, and complexity throughout the Austrian citizenship process. The dataset contains **17 structured steps** and models the full process from document preparation to final approval.

The analysis focused on four areas:

- the most time-consuming stages
- the most costly requirements
- the most complex steps
- the relationship between time and difficulty

Time analysis

The total process duration captured in the dataset is **240 days**. Python and SQL analysis both showed that the process is dominated by waiting periods rather than active administrative work.

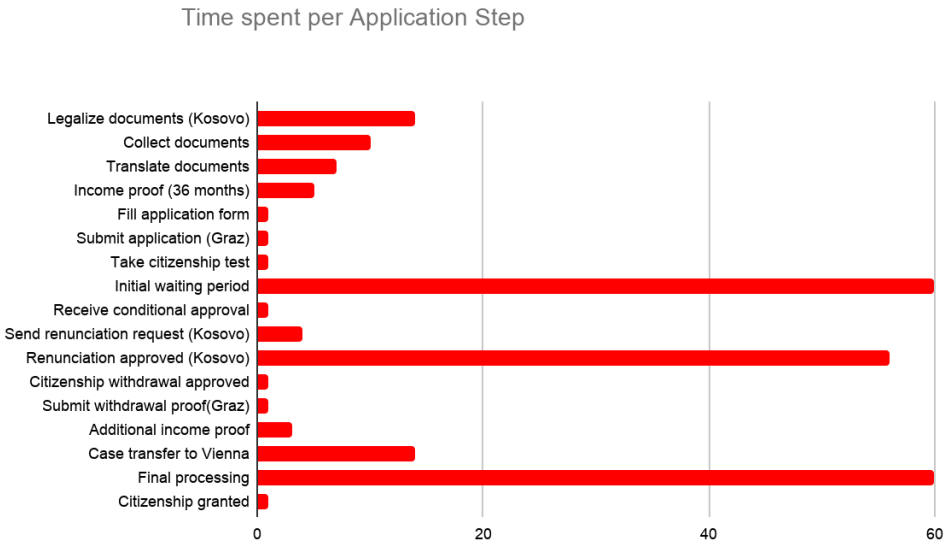
The longest individual steps are:

- **Wait for decision** — 60 days
- **Final processing period** — 60 days
- **Wait for Kosovo decision** — 56 days
- **Legalize documents from Kosovo** — 14 days
- **Case transferred to Vienna** — 14 days

At category level, the **Waiting** category accounts for **176 days**, which is by far the largest share of total process time. This makes waiting periods the dominant time burden in the entire process.

Interpretation:
The Austrian citizenship process is primarily time-intensive because of passive delays, many of which remain outside the applicant's control. The involvement of a second state system, Kosovo, further extends the process and increases uncertainty.

Figure 1:



Cost analysis

The total direct cost recorded in the dataset is **€1,740**. Costs are concentrated in a small number of steps, not spread evenly across the process.

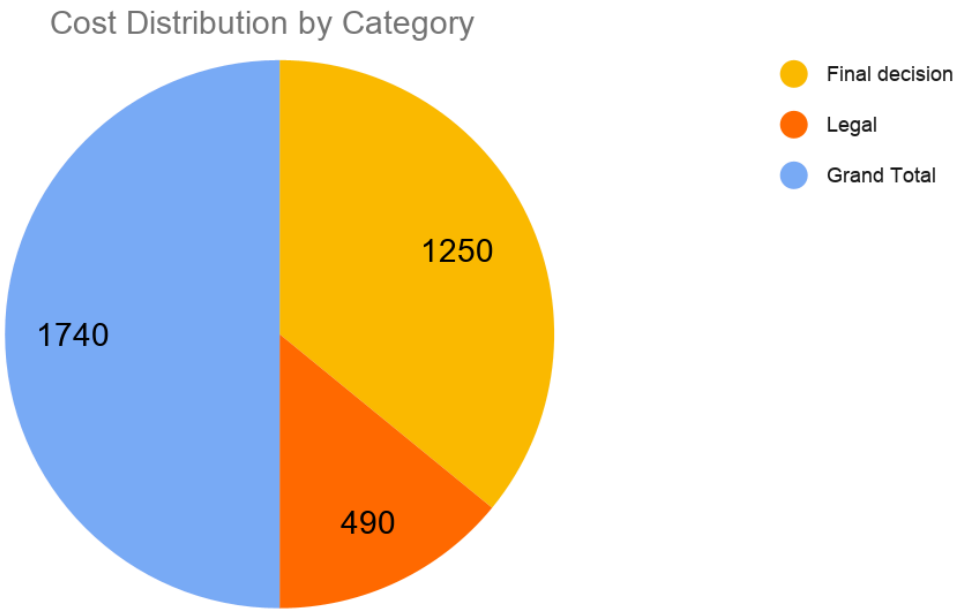
The main direct costs are:

- **Citizenship granted** — €1,250
- **Legalize documents from Kosovo** — 320
- **Send renunciation request (Kosovo)** — €120
- **Translate documents (certified)** — €50

At category level, cost is concentrated in:

- **Outcome / Final decision stage** — €1,250
- **Legal** — €490

Interpretation:
The majority of costs are associated with legal compliance and the final approval stage, rather than everyday administrative actions. This shows that the financial burden is created mainly by regulatory requirements and final status change, not by routine filing.



Difficulty analysis

The dataset rates difficulty on a scale of 1 to 5. The highest-difficulty steps are concentrated in areas that require legal interpretation, financial proof, and cross-border coordination.

The highest-rated steps are:

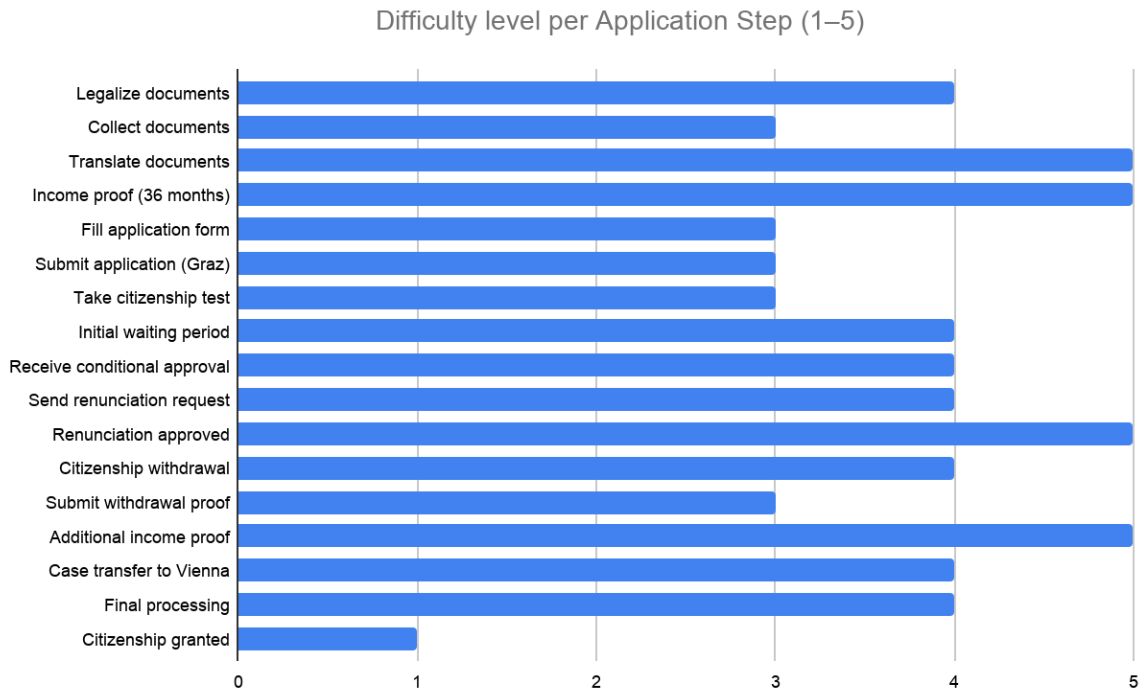
- legalize documents from Kosovo — 5
- translate documents (certified) — 5
- prepare income proof (36 months) — 5
- wait for Kosovo decision — 5
- provide additional income proof — 5

Lower-difficulty steps include:

- fill application form — 3
- submit application (Graz) — 3
- take citizenship test — 3
- submit withdrawal proof to Graz — 3
- citizenship granted — 1

Interpretation:

The most difficult parts of the process are not the simple filing steps. They are the stages where the applicant must interpret legal rules, assemble multi-year financial proof, and navigate external institutions.



Relationship between time and difficulty

One of the clearest findings in the dataset is that **time and difficulty do not perfectly overlap**.

Examples of high-difficulty but shorter steps:

- document legalization
- document translation
- income proof preparation

Examples of lower-effort but highly time-consuming steps:

- waiting for Austrian decisions
- waiting for Kosovo's citizenship withdrawal decision

Interpretation:
The total burden of the process comes from two different sources:

1. **cognitive complexity** — understanding rules, requirements, and documentation
2. **system-driven delay** — waiting periods and cross-border administrative dependency

Summary of analysis findings

The analysis shows that:

- waiting periods account for the majority of the total duration

- legal and final-decision stages account for most of the cost
- the most difficult steps are legal, financial, and cross-border in nature
- time and difficulty are related, but not identical
- external dependency on Kosovo significantly increases duration and complexity

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5. Share

The story told by the data is clear: the Austrian citizenship process is not burdensome only because it is strict, but because it combines **extended waiting periods**, **high-stakes legal requirements**, and **multi-system coordination**.

The visualizations support three high-level insights:

1. **Time is concentrated in waiting periods** rather than active work
2. **Cost is concentrated in a few legally significant steps**
3. **Difficulty is concentrated in legal, financial, and transnational requirements**

6. Act

Final conclusion

The Austrian citizenship process, when completed independently, is shaped by three dominant burdens:

- prolonged waiting time
- concentrated legal and final-stage costs
- procedural complexity tied to law, documentation, and cross-border bureaucracy

Top three recommendations

1. **Create a centralized digital guidance platform**
   Applicants should have access to one official, step-by-step digital system that clearly explains required documents, deadlines, sequencing, and status.
2. **Improve transparency for time and cost expectations**
   Authorities should provide clearer estimates for typical waiting times, document-related costs, and follow-up requirements so applicants can plan more accurately.
3. **Provide clearer support for legal and financial documentation**
   The most difficult stages involve proving eligibility, especially income and foreign-document compliance. Better structured examples, templates, and explanations would significantly reduce applicant confusion.

Broader policy reflection

A broader reform worth considering is the expansion of **dual citizenship** access. Based on this case, renouncing prior citizenship adds both time and procedural burden while also raising questions of identity and continuity. Allowing applicants to retain their original nationality would reduce administrative friction and support a more inclusive naturalization process.

Next steps

If this project were expanded further, the next useful step would be to collect similar data from multiple applicants in order to compare experiences across:

- different regions in Austria
- different countries of origin
- different legal pathways to citizenship

That would make it possible to distinguish individual experience from broader system-wide patterns.

7. References

Referat Staatsbürgerschaft/Landesregierung Steiermark. Austrian Federal Government. (n.d.). Citizenship application requirements and document checklist.

Self-collected dataset based on personal application experience.